

Numerical validation of `finite_contour_cocycles_short.tex`

Companion to `validate.py`

1 What this suite is for

Every closed form in the note is a claim that some explicitly-summable expression equals a contour integral. Algebra alone does not catch a claim that is right in one geometric regime and wrong in another: the error is invisible until the wrong regime is evaluated. This suite therefore takes *raw quadrature of the defining contour integral* as ground truth and checks each closed form against it, never against another closed form.

Two design rules follow.

Index by regime, not by equation. Each check runs across the geometric configurations the note distinguishes: a pole above the T -segment, on the edge $\text{Im } z = 1$ with $z \neq i$, at i , below the contour, and absent. A per-equation test that samples one configuration certifies nothing about the others.

Ground truth must be exact. Where a check needs Fourier coefficients, they come from integer q -series arithmetic, not from numerical extraction. Recovering $c_f(n)$ from $\int_0^1 f(x+iy)e^{-2\pi in(x+iy)}dx$ is catastrophically ill-conditioned for a form with a pole at the cusp: for $1/\Delta$ the integrand runs as $e^{2\pi ny}$ while the answer is only $e^{4\pi\sqrt{n}}$, so $c(18)$ at $y = 1.3$ asks for a 10^{23} number out of 10^{67} -sized values. The suite computes $\prod(1 - q^n)^{\pm 24}$ in exact integer arithmetic instead; `cf_Delta` and `cf_invDelta` reproduce $\tau(n)$ and the known $1/\Delta = q^{-1} + 24 + 324q + 3200q^2 + \dots$ exactly.

Each check carries the `\label` it validates, so a failure names a line in the note rather than a formula in Python. Two tiers: `validate.py` (coarse precision, one case per regime) and `validate.py --slow` (high precision, full cross-product). A third tier, `report`, prints a measured discrepancy without passing or failing; it is used for questions the note has not yet settled, so the suite documents them instead of presuming an answer.

Shared machinery in validate.py

<code>ktil(tau,s,k)</code>	the Hurwitz T -kernel $\tilde{k}_T$ of <code>def:kernel</code>
<code>LT_raw(f,k,s,tau0)</code>	ground truth: quadrature of $e^{-i\pi s/2} \int_{\tau_0-1}^{\tau_0} f \tilde{k}_T d\tau$
<code>LT_closed(cf,k,s,tau0)</code>	the closed form <code>eq:T-segment-closed-Mk</code>
<code>E4, E6, Delta, invDelta</code>	the forms as functions on $\mathbb{H}$, for quadrature
<code>cf_E4, cf_E6, cf_Delta</code>	their <i>exact</i> integer q -coefficients
<code>cf_invDelta, cf_j, jay</code>	likewise for $1/\Delta$ and $j = E_4^3/\Delta$
<code>residue_at(g,s0,r)</code>	Res by trapezoid rule on a circle $ s - s_0 = r$
<code>WEAK_FORMS</code>	the Layer C form table (form, function, coefficients, k)
<code>_eta_prod, _smul, _sinv</code>	integer power-series arithmetic behind those
<code>check(label,layer,tier)</code>	decorator registering a test against a <code>\label</code>
<code>rel(got,want)</code>	relative error, denominator floored at 1
<code>main</code>	driver; TOL, NT, NMAX are set by tier

2 Layer A: kernel and unfolding

Layer A sits upstream of everything else. If the kernel identities or the mode-by-mode unfolding are wrong, every downstream check is measuring the wrong thing.

A1 — the δ_T shift (`def:kernel; a1_kernel_shift`). Verifies $\delta_T \tilde{k}_T(\tau, s) = \tilde{k}_T(\tau + 1, s) - \tilde{k}_T(\tau, s) = -(\tau + 1)^{s-1} + e^{i\pi(s-1)}(\tau + 1)^{k-1-s}$ directly, at three weights, three base-points and three values of s including one on the critical line. This is the identity that turns the ∂_{τ_0} boundary term into a unit shift of the kernel, so it underwrites τ_0 -independence.

A2 — the Hurwitz antiderivative (`eq:hurwitz-antideriv; a2_hurwitz`). Three separate claims: $\partial_a [\zeta(\sigma - 1, a)/(1 - \sigma)] = \zeta(\sigma, a)$; the shift $\zeta(\sigma - 1, a + 1) - \zeta(\sigma - 1, a) = -a^{1-\sigma}$; and the definite integral $\int_{i-1}^i \zeta(\sigma, \tau + 1) d\tau = i^{1-\sigma}/(\sigma - 1)$ that they combine to give. These produce the $n = 0$ contribution, hence the entire $-c_f(0)(\frac{1}{s} + \frac{i^k}{k-s})$ polar structure.

A3 — the unfolding (`eq:T-segment-closed-Mk`). The general- τ_0 T -segment closed form, checked against quadrature in three stages of increasing exposure:

- `a3a_unfold_pos` — positive modes only ($\Delta, k = 12$): no branch question arises;
- `a3b_unfold_const` — $c_f(0) \neq 0$ ($E_4, k = 4$): exercises the $-c_f(0)((\tau_0/i)^s/s + \dots)$ term;
- `a3c_unfold_neg` — negative modes ($1/\Delta, k = -12$): $n < 0$ makes $(2\pi n)^s$ a power of a negative number and drives $-2\pi i n \tau_0$ onto the cut of $\Gamma(s, \cdot)$.

The third stage is the “incomplete-gamma unfolding ambiguity”. The suite pins the convention by quadrature: for $n < 0$,

$$(2\pi n)^s := (2\pi |n|)^s e^{+i\pi s}.$$

This must be applied as an explicit phase and never folded into the base as $(2\pi |n| e^{i\pi})^s$: numerically $e^{i\pi} = -1 + \varepsilon i$ with the sign of ε a rounding artifact, so the base-folded form silently changes branch with the working precision. With the phase explicit, the closed form matches quadrature to 10^{-31} at $\tau_0 = i, 1.35i, 2i$ and at τ_0 off-axis to the right.

A3, $\text{Re } \tau_0 < 0$: the continued Γ (a3d_unfold_neg_left). For $n < 0$ write $z = -2\pi i n \tau_0$. With $\tau_0 = x + iy$ this is $z = -2\pi |n|y + 2\pi i |n|x$, so $\text{Re } z < 0$ always while $\text{Im } z$ carries the sign of $x = \text{Re } \tau_0$: the point z sits just above the negative-real cut of $\Gamma(a, \cdot)$ when $\text{Re } \tau_0 > 0$ and just below it when $\text{Re } \tau_0 < 0$. The L-integral varies continuously as τ_0 moves; the *principal* Γ jumps.

The jump is the standard monodromy. Since $\Gamma(a, z) = \Gamma(a) - \gamma(a, z)$ with $\gamma(a, z) = z^a \sum_{m \geq 0} (-z)^m / (m!(a+m))$, all the multivaluedness sits in z^a , and one anticlockwise turn gives

$$\Gamma_{\text{cont}}(a, z) = \Gamma(a)(1 - e^{2\pi i a}) + e^{2\pi i a} \Gamma_{\text{prin}}(a, z),$$

implemented as `_gamma_cont` and applied per term at $a = s$ and $a = k - s$. With it, `eq:T-segment-closed-Mk` reproduces quadrature to 10^{-31} for $\text{Re } \tau_0 < 0$, just as the principal branch does for $\text{Re } \tau_0 \geq 0$.

So the lemma is correct as stated for all $\tau_0 \in \mathbb{H}$, and `prop:HGG`'s phrase “the identity continues to $n < 0$ ” is doing real work: Γ must be the continuation, which coincides with the principal value only while $\text{Re } \tau_0 \geq 0$. An earlier draft of this document reported the lemma as failing for $\text{Re } \tau_0 < 0$; that was an artifact of evaluating Γ on the principal branch on the far side of its cut, not a defect of the result.

3 Layer B: τ_0 -independence and wall-crossing

Layer B tests the full L-integral $L^* = L_S^* + L_T^*$ of `def:Lint`, with both segments by quadrature (`LS_raw`, `LT_raw_path`, assembled by `L_raw`). Paths are polylines integrated by `path_int`; $\mathbb{H}$ is convex, so the straight line from $-1/\tau_0$ to τ_0 stays inside it.

B1, B2 — τ_0 -independence (b1_indep_onaxis, b2_indep_offaxis). `prop:indep`: neither segment is separately invariant, only the sum. Checked on the imaginary axis and off it, for Δ , E_4 , E_6 and $1/\Delta$. B2 includes $\text{Re } \tau_0 < 0$ deliberately, which is what first showed that the A3 discrepancy could not be a property of the L-integral: the raw L^* is invariant there to 10^{-16} , so whatever was wrong had to lie in the evaluation of the closed form — as it did.

B3 — a meromorphic form (b3_indep_meromorphic). `prop:indepHomotopy` for $f = \Delta/(j - j(2i)) \in F_{12}$, whose poles are the Γ -orbit of $2i$ at heights $2, \frac{1}{2}, \frac{2}{5}, \dots$. Base-points near i keep both segments clear of every image, so these all lie in one class and L^* must not move.

B4, B5 — wall-crossing (b4_wall_T, b5_wall_S). `lem:wall` with *fixed end-points and two homotopy classes*, so the crossing is isolated from the genuine τ_0 -dependence of each separate segment. For T : a straight segment at height 2.2 against one dipping to 1.8, enclosing $2i$ alone. For S : two paths agreeing around every image except $2i$, which one passes to the left and the other to the right. In both the enclosed loop runs clockwise, so $X = -1$ and the predicted jump is $-2\pi i e^{-i\pi s/2} r$, with $r_T = \text{Res}(f \tilde{k}_T)$ and $r_S = \text{Res}(f \tau^{s-1})$ from `res_circle`.

Two things had to be got right for these to test anything. The form is normalised so $\text{Res}_{2i} f = 1$; unnormalised the residue is $\Delta(2i)/j'(2i) \approx 2 \times 10^{-12}$ and every jump is $\sim 10^{-9}$, at which scale a sign error hides under the noise floor. And `res_circle` needs enough points: the trapezoid rule on $|\tau - 2i| = r$ converges like $(r/R)^M$ with $R = 1$ the distance to $2i + 1$, so the original $r = 0.3$, $M = 16$ gave only $0.3^{16} \sim 4 \times 10^{-9}$ — agreement to eight digits, which reads as a failure against a 10^{-9} tolerance and invites a hunt for a nonexistent bug elsewhere.

4 Layer C: `thm:weakL` and the functional equation

For $f \in M_k^!$ the base-point may be taken at $\tau_0 = i$, where the S -segment degenerates to a point and $L^* = L_T^*$ alone. Ground truth for this layer is therefore `LT_raw(f,k,s,I)`, the bare quadrature of $e^{-i\pi s/2} \int_{i-1}^i f \tilde{k}_T d\tau$.

The five test forms (`WEAK_FORMS`) are chosen to span the cases the statement distinguishes rather than to be interesting individually:

form	code	k	feature exercised
Δ	<code>cf_Delta</code>	12	cuspidal form, $c_f(0) = 0$
E_4	<code>cf_E4</code>	4	$c_f(0) \neq 0$, so the polar term is live
E_6	<code>cf_E6</code>	6	$i^k = -1$
$1/\Delta$	<code>cf_invDelta</code>	-12	pole at the cusp, so $n < 0$ modes
j	<code>cf_j</code>	0	the $s = 0$ and $s = k$ poles coincide

$j = E_4^3/\Delta$ is built by integer series arithmetic and reproduces $q^{-1} + 744 + 196884q + 21493760q^2$ exactly; E_6 reproduces $1 - 504q - 16632q^2 - 122976q^3$.

C1 — `eq:weakL` (`c1_weakL`). The closed form against quadrature, for all five forms. Since `eq:weakL` is `eq:T-segment-closed-Mk` specialised to $\tau_0 = i$, this also re-exercises the branch convention pinned in A3, now with $c_f(0) \neq 0$ and with $k \leq 0$.

C2 — the residue (`c2_residue`). $\text{Res}_{s=0} L^* = -c_f(0)$. The residue is extracted from the *raw* integral by a contour in s (`residue_at`, trapezoid rule on $|s| = r$), so the note's polar claim is tested against quadrature and not against its own closed form. The circle is used rather than extrapolating $s L^*(s)$, which converges only linearly in s and would not reach tolerance.

The weight-zero case is the sharp one. For $k = 0$ the two poles collide, and the polar term is $-c_f(0)(\frac{1}{s} + \frac{i^0}{0-s}) = 0$ identically, so the residue must *vanish* even though $c_j(0) = 744$. It does. This is the mechanism behind the factor $(1 - \delta_{k,0})$ appearing in the special-value corollaries of arXiv:1806.09874, recovered here from bare quadrature.

C3 — the functional equation (`c3_functional_eq`). $L^*(f, s) = i^k L^*(f, k - s)$, with *both* sides computed by raw quadrature. This is the cheapest global probe in the suite: it touches every phase in the construction at once, and a single lost sign anywhere breaks it. E_6 is the case that matters, being the only form here with $i^k = -1$; the weights 12, 4, -12, 0 all give $i^k = +1$ and would not detect a dropped i^k .

5 Layer D: periods and quasi-periods

Two of the three checks here are *external*: they compare the note's construction against classical objects computed by an independent route, rather than against another of the note's own formulas. The test forms are the unique normalised cusp forms $\Delta_k = q + O(q^2)$ at the weights with $\dim S_k = 1$ (`CUSP1`: $k = 12, 16, 18, 20, 22, 26$), built by exact integer series multiplication from Δ , E_4 , E_6 ; $\Delta_{16} = \Delta E_4$ reproduces $q + 216q^2 - 3348q^3 + \dots$ exactly.

D1 — $r_f \in W$ (`d1_rfW`). `def:rf` assembles the period polynomial from the L^* -values, $r_f = (2\pi i)^{n+1} \sum_{\ell} (-1)^{\ell} \binom{n}{\ell} i^{\ell+1} L^*(f, \ell+1) X^{n-\ell} Y^{\ell}$, and `lem:rfW` claims it lies in W . The test applies the slash action $(w|\gamma)(X, Y) = w(aX + bY, cX + dY)$ (`slash`) at S and $U = TS$, and checks $r_f|_{(1+S)}$ and $r_f|_{(1+U+U^2)}$ vanish relative to the size of r_f . They do: the S -relation to machine zero, the U -relation to 10^{-11} – 10^{-13} . So the period polynomial built from the finite two-segment contour satisfies the Eichler–Shimura relations.

D2 — **the classical periods** (`d2_classical_periods`). **External anchor.** $r_m = i^{m+1} L^*(m+1)$ must be the classical critical period $\int_0^{i\infty} \tau^m f(\tau) d\tau = i^{m+1} \int_0^{\infty} y^m f(iy) dy$, i.e. L^* must be the completed period $\Lambda(f, s)$. The right side is obtained by direct quadrature on the axis, not by re-deriving the incomplete-gamma sum, so this fixes the normalisation of `def:rf` independently. Agreement is 10^{-11} – 10^{-16} at both ends of the critical strip.

The integrand must be evaluated through `f_on_axis`. A cusp form decays at both ends of the axis, but the q -series does not converge at the lower one: at $y = 0.03$ one has $|q| = 0.83$, where forty terms of E_4 are meaningless. Below $y = 1$ the evaluation is pulled back by $f(iy) = f(i/y)/(i^k y^k)$, which lands it above the unit circle. Without that the computed period comes out as 6×10^{11} in place of 10^{-73} .

D3 — **Haberland–Petersson** (`d3_haberland`). **The strongest check in the suite**, because it is normalisation-*invariant* and both sides are computed here from scratch. Cohen, ANTS X (2013), Thm 5.2(2), for the full modular group:

$$-6(-2i)^{k-2} \langle f, f \rangle = \sum_{m+n \leq k-2} \binom{k-2}{m+n} \binom{m+n}{m} (-1)^m \operatorname{Im}(r_m \overline{r_n}),$$

with $r_m = i^{m+1} L^*(f, m+1)$. The left side is the Petersson norm by direct integration of $|f|^2 y^{k-2}$ over the fundamental domain; the right uses only the finite-contour L^* . They agree to 6×10^{-16} .

Unlike the determinant ratios of Table 1, which depend on how the period-polynomial basis is normalised, nothing here can be rescued or spoiled by a choice of basis: the pairing is contracted away on both sides. That makes it the closest thing in this suite to an outside verdict on the construction.

6 Layer E: the meromorphic machinery of §4

This is the layer with the most apparatus and the least external corroboration, so it carries two ground truths of its own. `IN_raw` evaluates $I_N(s, z)$ directly: by `prop:indepHomotopy` the $\delta \rightarrow 0$ representative of the minimal class may be used, and there the block is regular at i (as $\operatorname{Im} z > 1$), so the S -integral vanishes and the T -segment is $[i-1, i]$. `wallvalue_raw` evaluates $L^*(\hat{P}_i f, s; i(1+\delta))$ at finite δ , the S -segment split into a regular part and the principal value of the pure pole, as `def:null_homotopy` prescribes. Three meromorphic forms with a pole at i span the pole orders (`MERO_AT_I`): Δ/E_6 ($k=6, P=1$), $E_4^2 \Delta/E_6^2$ ($k=8, P=2$), Δ^2/E_6^3 ($k=6, P=3$) — orders and weights consistent with `prop:ellord`.

E1 — **the Γ recursion** (`e1_gammarec`). `eq:gammarec` at three cutoffs J , three arguments X (real, complex, large) and two s . Cheap, and upstream of both `lem:polylogedge` and `lem:wallvalue`.

E2 — the block expansion (e2.blockexp). `eq:blockexp` claims the block is a pure order- m pole plus an entire T -periodic tail. The test forms the remainder $\frac{(-2\pi i)^m}{(m-1)!} \text{Li}_{1-m}(e^{2\pi i(\tau-i)}) - (\tau-i)^{-m}$ and checks its residue at i vanishes, for $m = 1, 2, 3$. That vanishing is precisely what makes $r_{f,z}^*(m)$ the Laurent coefficients rather than merely proportional to them, so it validates the Option-A reading of `def:proj`.

E3 — I_N above the line (e3.polylogcf). `lem:polylogcf` against `IN_raw`, at $N = 0, 1, 2$ and z both on and off the imaginary axis. Everything in §4 is built on this lemma, and $N = 0$ is the case carrying the $\delta_{N,0}(\frac{1}{s} + \frac{i^k}{k-s})$ inversion constant — the term responsible for the $1/s$ poles that distinguish this L^* from the cusp-anchored one of arXiv:1806.09874. It reproduces quadrature to 10^{-16} .

E4 — the wall value (e4.wallvalue). `lem:wallvalue` asserts a *limit*, so the raw integral must be extrapolated rather than sampled: $L = (8f(\delta/4) - 6f(\delta/2) + f(\delta))/3$ annihilates both an $O(\delta)$ and an $O(\delta^2)$ error term. Both orders occur, and which one appears is not incidental — it splits along the parity dichotomy of `prop:ellord`:

form	regime	convergence
$\overline{E}_4^2 \Delta / E_6^2$	$4 \mid k, P \text{ even}$	$O(\delta^2)$, ratio $\rightarrow 4$ per halving
$\Delta / E_6, \Delta^2 / E_6^3$	$k \equiv 2 \pmod{4}, P \text{ odd}$	$O(\delta)$, ratio $\rightarrow 2$

All three converge to $\sum_m \frac{(-2\pi i)^m}{(m-1)!} r_{f,i}^*(m) \text{FP}_{m-1}(s, k)$. The tolerance here is 10^{-5} , not the global one: what survives extrapolation is the $O(\delta^3)$ term the three-point formula cannot remove. The substantive evidence is the convergence *rate*, not the residual.

E5 — the cutoff (e5.cutoff, e5b.cutoff_minimal). `rem:cutoff` holds that FP_N is independent of J for $J \geq N + 1$. It is, but each J carries its own truncation: the remainder decays as $n^{-(J-N+1)}$, so at a fixed $n_{\max}$ only the larger cutoffs are numerically converged. The equality is asserted among those; the slower ones, including the minimal $J = N + 1$, are shown converging in a `report-tier` table — at $J = N + 1$ the error falls as $1/n_{\max}$, tripling $n_{\max}$ thirding the error, exactly the n^{-2} tail the note predicts. `FP` therefore *evaluates* at $J = N + 5$, which is legitimate precisely because the lemma being tested is true.

E6 — the modularity relations (e6.laurent). `eq:Laurent` at every order $r = 1, \dots, P$ for all three forms; holds to machine zero.

E7 — the edge route (e7.edge_route, report tier). A pole at $\text{Im } z = 1$ with $z \neq i$ admits two readings. `lem:polylogedge` continues the from-*above* formula of `lem:polylogcf` down to the edge, which is the convention the note adopts and retains. The from-*below* value — the block's own q -series against $G_{s,k}$, with no inversion and hence no $\delta_{N,0}$ term — differs from it by the crossing residue, measured here as a factor of about 4.4 at $z = 0.3 + i$. The line is printed rather than scored: it records the size of a convention choice, not a defect.

A note on tolerances. Two Layer E checks carry their own tolerance instead of the global TOL. This is deliberate. TOL scales with the working precision, but a check limited by truncation or by extrapolation does not improve when mpmath is handed more digits: at $\text{dps} = 30$ the global tolerance is 10^{-24} , while $\mathcal{E}^+$ truncated at $n_{\max} = 60$ caps agreement near 10^{-15} however many digits are carried. A tolerance must reflect the numerics of the test that uses it, not the width of the mantissa.

7 Layer F: external anchors

Layers A–E check the note against itself and against raw quadrature. Layer F checks it against things outside it: the earlier paper, and a statement about how the two must differ.

F1 — the published $y < B$ branch (f1_1806_below). eqLemRegX3 of arXiv:1806.09874 in the case where the ray never crosses the circle $|e(it - \tau_p)| = 1$. It is correct as published, reproducing I1806_raw to 10^{-16} for $N = 0, 1, 2$. It also pins that paper’s conventions: the integrand is the proof’s $\frac{dt}{t}t^s$, not the definition’s $\frac{d\tau}{\tau}(\tau/i)^{s-1}$.

F2 — the corrected $y > B$ branch (f2_1806_above). Regression test for the corrigendum in fix_previous_1806_file/. For $y > B$ the ray crosses the circle at $t = y$, and neither the q -series nor its inversion is valid across that point, so the integral must be split there. The published version evaluates a single antiderivative at $t = B$ and $t = \infty$, which drops three things: the $t = y$ end-point, the $\delta_{N,0}$ constant from the $N = 0$ inversion, and the whole $\int_y^\infty$ tail. The corrected form reproduces quadrature to 10^{-6} ; the published one misses by a factor of about 12.

Two implementation notes, both of which cost a wrong answer before being fixed. The branch is the one the corrigendum specifies by $[\Gamma(s, -2\pi nB) - \Gamma(s, -2\pi ny)]/(-2\pi n)^s = \int_B^y \frac{dt}{t}t^s e^{2\pi nt}$; with $(-2\pi n)^s$ read principally as $(2\pi n)^s e^{+i\pi s}$, dividing contributes $e^{-i\pi s}$, which matches that integral to 10^{-25} . Pairing $e^{+i\pi s}$ with mpmath’s default branch instead converges to a value 26% off. And the two boundary series converge only conditionally — their terms go as n^{N-1} times a phase — so partial sums are Cesàro-averaged, which is why the tolerance here is 10^{-4} rather than the global one.

F3 — entirety of the 1806 block integrals (f3_1806_entire). Thm 1 of that paper asserts the only non-regular terms are $-c_f(0)(\frac{B^s}{s} + \frac{i^k B^{k-s}}{k-s})$. That needs the block integrals to be regular in s , which is automatic — a finite lower end-point plus exponential decay at the cusp makes $\int_B^\infty$ converge for every s — and is checked here as a residue in s taken around the raw integral. It vanishes to 10^{-17} . This is why the defect in F2, which is a finite-part error, cannot disturb Thm 1, the Hurwitz class-number sum rule, or any of that paper’s special values: all of those are residue statements, and $1/\Gamma(s)$ annihilates every regular term at $s = 0$.

F4 — the residue-strip rule (f4_residue_strip). The sharp form of how this L^* and the cusp-anchored one differ. L_S^* is regular in s , so the pole at $s = 0$ comes entirely from the T -segment, where the kernel’s Hurwitz pole $\tilde{k}_T \sim -1/s$ multiplies the constant Fourier mode. Hence

$$\text{Res}_{s=0} L^*(f, s) = - \int_{\tau_0-1}^{\tau_0} f(\tau) d\tau,$$

the integral over one horizontal period being exactly that constant mode — read in *the strip the contour sits in*, not at the cusp. For $\Delta/(j - j(2i))$, whose pole at $2i$ lies above the contour, this holds to 2×10^{-11} ; for E_4 , E_6 and $1/\Delta$ to 10^{-14} .

This is what dissolves the apparent conflict with arXiv:1806.09874. That paper anchors its contour at the cusp, above every pole, so it sees only $c_f(0)$; this one sits at height one and additionally sees $2\pi i \sum a_{-1}$ from the poles above it. Both are right about their own object, and the difference is exactly the residues crossed in moving the base-point. A cusp form is excluded from the check: there both sides vanish identically and a relative comparison carries no information.

F5 — the Hurwitz class numbers and traces (f5_hurwitz_qexp). **External arithmetic anchor.** For class number one the Hilbert class polynomial is linear, so $\Lambda_d = -q \partial_q \log \mathcal{H}_d(j(\tau))$ is elementary:

$$\Lambda_3 = \frac{E_6}{3E_4}, \quad \Lambda_7 = \frac{E_4^2 E_6}{E_4^3 + 3375 \Delta},$$

using $q \partial_q j = -E_4^2 E_6 / \Delta$ and $j + 3375 = (E_4^3 + 3375 \Delta) / \Delta$, each normalised so that $c(0) = H(d)$. Read off at height 2.2, above every pole, the q -expansion must be $\Lambda_d = H(d) + \sum_n t_n(d) q^n$ with $H(d)$ the Hurwitz class number and $t_n(d)$ the trace of J_n over CM points of discriminant $-d$:

	$c(0)$	$H(d)$	$c(1)$	$t_1(d)$
Λ_3	0.333333333333	1/3	-248.0	$(j(\rho) - 744)/3 = -248$
Λ_7	1.0	1	-4119.0	$j(\alpha_7) - 744 = -4119$

Three of the four agree to exactly zero. This is Zagier's arithmetic recovered by contour integration, and it is the one place where the construction is measured against mathematics that predates it.

F6 — the Hurwitz sum rule (f6_hurwitz_sumrule). Eq. (s3e2) of arXiv:1806.09874 states $\lim_{s \rightarrow 0} L_{\Lambda_d}^{\text{reg}}(s) = -H(d)$. Since $L^{\text{reg}} = (2\pi)^s L^*/\Gamma(s)$ and $1/\Gamma(s)$ has a *zero* at $s = 0$, the limit picks out the residue alone, so by F4 it equals $-c_f(0)$ read in the strip the contour sits in. The two discriminants then behave completely differently:

- $d = 3$: the orbit of ρ has maximal height $\sqrt{3}/2 < 1$, so every pole lies below the contour, the strip constant *is* the cusp constant, and the residue is $-H(3) = -1/3$ — the sum rule exactly as published.
- $d = 7$: the CM point $\alpha_7 = (1 + i\sqrt{7})/2$ has height $\sqrt{7}/2 > 1$ and lies *above* the contour. A simple zero of $j - j(\alpha)$ always carries $a_{-1} = -1/(2\pi i)$, so it contributes $2\pi i a_{-1} = -1$ to the strip constant, which cancels $H(7) = 1$ exactly. The residue **vanishes**, to 1.4×10^{-16} .

Same form, same machinery, and the $1/s$ residue is $-H(d)$ or 0 purely according to which side of the contour the CM point falls on. This is the entire difference between this L^* and the cusp-anchored one of arXiv:1806.09874, reduced to a single number — and it is why the apparent conflict between the two was never a contradiction.

F7 — lifting the contour above every pole (f7_raised_contour). The test behind `rem:raised`. Since L_S^* is a finite integral of $f \tau^{s-1}$ and is entire in s , the pole at $s = 0$ is carried entirely by the T -segment, where $\tilde{k}_T \sim -1/s$ multiplies $\int_{\hat{\gamma}_f^T} f d\tau$. Deforming $\hat{\gamma}_f^T$ to pass above every pole of f should therefore return the constant term at the *cusp*, recovering the residue bookkeeping of arXiv:1806.09874. Two cases, chosen so that lifting matters in one and not the other:

	poles vs. contour	flat Res	lifted Res
Λ_7	α_7 at height $\sqrt{7}/2 > 1$, above	0	$-1 = -H(7)$
Λ_3	orbit of ρ at $\sqrt{3}/2 < 1$, all below	$-1/3$	$-1/3 = -H(3)$

In both the lifted value is $-c_f(0)$ at the cusp. For Λ_7 that is a genuine change of class, and the shift is the crossing of Lemma ??; for Λ_3 nothing is crossed and the lift is a homotopy, so the value must not move, which it does not. The base-points are chosen so the vertical legs of the lifted path miss the poles: the orbits of α_7 and of ρ both sit at $\text{Re} = \frac{1}{2} \bmod 1$, so legs at $\text{Re} = -1, 0$ are clear.

A form whose constant term vanishes at the cusp is useless for this check. $\Delta/(j - j(2i))$ begins at q^2 , so both sides are zero and the comparison carries no information; the same consideration excludes cusp forms from F4.

8 Layer G: suite hygiene

The suite is the arbiter, so it is itself checked. Each of these recomputes a quantity with one numerical knob changed and demands the answer not move.

G1 — truncation (g1_truncation). Mode sums under $n_{\max} \rightarrow 2n_{\max}$, for L^* and for FP_N . This is the failure that made FP at the minimal cutoff $J = N+1$ look like a violation of `rem:cutoff`.

G2 — precision (g2_precision). Values recomputed at $\text{dps} + 12$. This catches results that are artifacts of the mantissa rather than of the mathematics — the failure mode behind the branch-in-base bug of A3, whose answer flipped between $\text{dps} = 15$ and 25 because $e^{i\pi} = -1 + \varepsilon i$ with the sign of ε a rounding artifact.

G3 — quadrature (g3_quadrature). Path integrals under refinement of the panel count, and `res_circle` under a change of radius. This is the knob that made the `lem:wall` checks miss tolerance while agreeing to eight digits.

Provenance

Over the sessions that produced this suite it found one defect in published work — `LemRegX3` of arXiv:1806.09874, corrected in `fix_previous_1806_file/` — and none in `finite_contour_cocycles_short.tex`. Everything else it turned up was a defect in the tests themselves: ill-conditioned coefficient extraction, a branch folded into a base, a relative-error floor, a winding orientation, an under-resolved residue, a sampled limit, a q -series evaluated outside its disc of convergence, and a patch that silently failed to apply. That ratio is the argument for keeping it: the machinery under test survived, and the apparatus built to doubt it did not.